Project Report: Versatile License Plate Deblurring

Course: Introduction to Computer Vision (Spring 2026) **Project:** Project 5 (Difficulty: Challenging)

1. Introduction

The objective of this project is to develop a comprehensive pipeline to restore readability to license plates degraded by various types of blur. Motion blur and defocus (out-of-focus) blur are common real-world artifacts that hinder Automatic Number Plate Recognition (ANPR) systems. This project utilizes frequency-domain deconvolution techniques, specifically the Wiener filter, to reverse these effects.

2. Methodological Approach

The core of the project is based on the mathematical degradation model: $g(x,y) = f(x,y) * h(x,y) + \eta(x,y)$, where g is the captured image, f is the sharp scene, h is the Point Spread Function (PSF), and η is additive noise.

A. Versatile Simulation Pipeline

Our simulation pipeline is designed to test the mathematical validity of the restoration for different scenarios:

- **Blur Selection**: The system allows toggling between synthetic **motion blur** (modeled as a linear PSF) and **defocus blur** (modeled as a circular/disk PSF).
- **Controlled Degradation**: We apply the selected PSF and add Gaussian noise with a known variance to create a validation dataset.
- **Validation**: Because the exact degradation parameters are known, this pipeline serves as a benchmark for the deconvolution algorithm's accuracy.

B. Adaptive Real-World Restoration Pipeline

The real-world pipeline is optimized for images where degradation parameters are unknown:

- **Parameter Estimation**: The pipeline supports manual selection and tuning of the PSF type and its parameters (e.g., radius for defocus or length/angle for motion) based on visual inspection or frequency domain analysis.
- **Preprocessing**: To minimize edge-ringing artifacts during the Fast Fourier Transform (FFT), the pipeline includes a cropping step to isolate the license plate.
- **Wiener Restoration**: The system applies the Wiener filter using an estimated Noise-to-Signal Ratio (NSR), typically set as a constant λ.

3. Challenges and Solutions

A. Handling Multiple Blur Types

Different blurs require different PSFs. Motion blur requires estimating both displacement and direction, while defocus blur requires finding the correct radius of the circle of confusion. We solved this by implementing dedicated kernel generation functions for both linear and circular PSFs.

B. Noise Sensitivity in Deconvolution

Direct inverse filtering is highly sensitive to noise because it amplifies high frequencies where the PSF magnitude is low. We addressed this by implementing **Wiener Deconvolution**, which incorporates an NSR term to suppress noise amplification and provide a stable restoration.

C. Implementation of Frequency Domain Shifts

Performing deconvolution in the frequency domain requires padding the PSF to match the image size. This padding introduces a spatial shift in the restored image. We resolved this by calculating the necessary spatial roll (using np.roll) based on the original kernel dimensions to perfectly recenter the license plate.

4. Results and Evaluation

Performance was measured using the **Peak Signal-to-Noise Ratio (PSNR)** to compare restored images against their sharp originals in the simulation.

- **Simulation**: The restoration significantly improved PSNR values, typically recovering plates to a level where characters are clearly discernible even with high initial blur and noise.
- **Real-World**: For real-world images where a ground truth is unavailable, success was defined by the legibility of characters (e.g., successfully reading the "KE 52 99E" plate).

5. Course Concept References

The project integrates several key concepts from the course:

- **Convolution Theorem**: Converting spatial convolution into frequency-domain multiplication for efficient processing.
- **Point Spread Function (PSF)**: Modeling physical optical errors as mathematical operators.
- **Wiener Filter**: Applying statistical methods to balance image sharpening and noise suppression.
- **Noise Modeling**: Accounting for various sensor noise types, such as photon shot noise and readout noise, during the restoration process.